

Advanced Optimization

Chapter 4 — Trust-Region Methods

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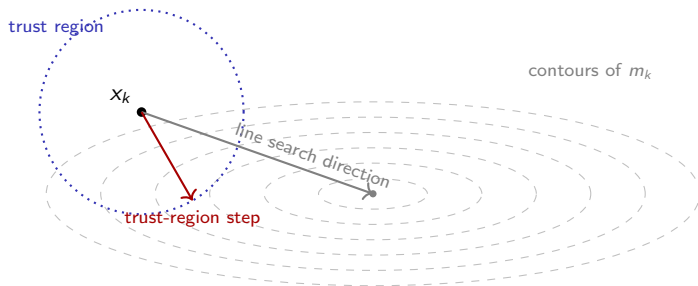
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Reading: Nocedal & Wright, *Numerical Optimization*, 2nd ed., Chapter 4 (and Algorithm 7.2 for CG–Steihaug)

Same model, opposite order (Figure 4.1)

Both strategies use a quadratic model of f around x_k . A line search uses it to pick a *direction* and then searches along it. A trust-region method picks a *region* in which it trusts the model, and minimizes the model there — direction and length at once.



If x_k and x^* sit at opposite ends of a curved valley, the line-search direction toward the model minimizer yields little; the trust-region step, bent by the bound, makes real progress. **The direction changes whenever the radius changes.**

The model and the subproblem

Taylor: $f(x_k + p) = f_k + g_k^T p + \frac{1}{2} p^T \nabla^2 f(x_k + tp) p$ (4.1), with $g_k = \nabla f(x_k)$. Replace the Hessian by a symmetric B_k :

$$m_k(p) = f_k + g_k^T p + \frac{1}{2} p^T B_k p. \quad (4.2)$$

Error $O(\|p\|^2)$; if $B_k = \nabla^2 f_k$, $O(\|p\|^3)$ — the **trust-region Newton** method (§4.4).

Trust-region subproblem

$$\min_{p \in \mathbb{R}^n} m_k(p) = f_k + g_k^T p + \frac{1}{2} p^T B_k p \quad \text{s.t.} \quad \|p\| \leq \Delta_k, \quad (4.3)$$

$\Delta_k > 0$ the **trust-region radius**; $\|\cdot\|$ Euclidean unless stated.

- ▶ Objective and constraint ($p^T p \leq \Delta_k^2$) are both quadratic.
- ▶ If $B_k \succ 0$ and $\|B_k^{-1} g_k\| \leq \Delta_k$: the solution is the **full step** $p_k^B = -B_k^{-1} g_k$. Otherwise, less obvious — but **an approximate solution suffices** for convergence and for good behavior.

Managing the radius: actual vs. predicted reduction

Given a step p_k , define

$$\rho_k = \frac{f(x_k) - f(x_k + p_k)}{m_k(0) - m_k(p_k)} = \frac{\text{actual reduction}}{\text{predicted reduction}}. \quad (4.4)$$

The denominator is always ≥ 0 ($p = 0$ is feasible). So:

- ▶ $\rho_k < 0$: f went *up*. Reject the step.
- ▶ $\rho_k \approx 1$: model and function agree over this step. Safe to *expand* the region.
- ▶ ρ_k positive but well below 1: keep the radius.
- ▶ ρ_k near 0 or negative: the model is unreliable at this size. *Shrink*.

Interpretation

The radius is a running estimate of *how far the model can be trusted*, updated from evidence. Consistently reliable model $\Rightarrow$ bolder steps; a failure $\Rightarrow$ retreat. This is the trust-region counterpart of the Wolfe conditions' role in a line search: the acceptance test.

Algorithm 4.1 (Trust Region)

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Given  $\hat{\Delta} > 0$ ,  $\Delta_0 \in (0, \hat{\Delta})$ ,  $\eta \in [0, \frac{1}{4}]$ :  
for  $k = 0, 1, 2, \dots$   
  Obtain  $p_k$  by (approximately) solving (4.3);  
  Evaluate  $\rho_k$  from (4.4);  
  if  $\rho_k < \frac{1}{4}$ :  $\Delta_{k+1} = \frac{1}{4}\Delta_k$   
  else if  $\rho_k > \frac{3}{4}$  and  $\|p_k\| = \Delta_k$ :  $\Delta_{k+1} = \min(2\Delta_k, \hat{\Delta})$   
  else:  $\Delta_{k+1} = \Delta_k$ ;  
  if  $\rho_k > \eta$ :  $x_{k+1} = x_k + p_k$    else:  $x_{k+1} = x_k$ .
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- ▶ $\hat{\Delta}$: an overall cap on step length.
- ▶ **The radius grows only if the step actually reached the boundary.** A step strictly inside means Δ_k was not the binding constraint — leave it.
- ▶ $\eta = 0$ accepts any decrease; $\eta > 0$ demands a fraction of the predicted decrease. The choice changes the global convergence statement (§4.2).
- ▶ The constants $\frac{1}{4}, \frac{3}{4}, 2$ are conventional, not sacred.

Everything else is about the one line “obtain p_k ”. Drop the subscript k from here on:

$$\min_p m(p) = f + g^T p + \frac{1}{2} p^T B p \quad \text{s.t. } \|p\| \leq \Delta. \quad (4.5)$$

What the exact solution looks like (Theorem 4.1, Moré–Sorensen)

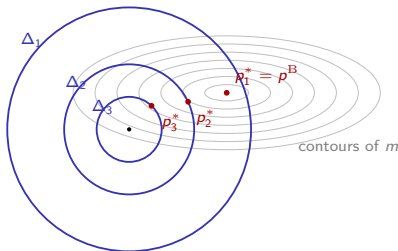
Theorem 4.1

p^* is a *global* solution of (4.5) if and only if p^* is feasible and there is $\lambda \geq 0$ with

$$(B + \lambda I)p^* = -g, \quad (4.8a)$$

$$\lambda(\Delta - \|p^*\|) = 0, \quad (4.8b)$$

$$B + \lambda I \succeq 0. \quad (4.8c)$$



Proof deferred to §4.3. Note the *global* — a nonconvex quadratic on a ball is one of the few nonconvex problems solved exactly.

Reading Theorem 4.1

- ▶ (4.8b) is a **complementarity** condition: at least one of λ and $\Delta - \|p^*\|$ is zero.
- ▶ Solution strictly inside (Δ_1 in the figure): $\lambda = 0$, so $Bp^* = -g$ with $B \succeq 0$ — the unconstrained minimizer.
- ▶ Solution on the boundary (Δ_2, Δ_3): $\lambda > 0$ allowed, and from (4.8a)

$$\lambda p^* = -Bp^* - g = -\nabla m(p^*),$$

so p^* is **collinear with the negative gradient of m** , normal to its contours.

- ▶ (4.8c): the multiplier is large enough to make $B + \lambda I$ positive semidefinite — **even if B itself is indefinite**. The trust region supplies its own Hessian modification, and its size is set by Δ , not by a heuristic.

Plan for the chapter

§4.1 cheap approximate solutions that do at least as well as the *Cauchy point* (dogleg, 2D subspace); §4.2 why that is enough for global convergence; §4.3 nearly exact solutions via Theorem 4.1; §4.4 fast local convergence; Ch. 7 the large-scale route (CG–Steihaug), previewed here.

The Cauchy point (Algorithm 4.2, Figure 4.3)

As with line searches, the optimal subproblem solution is not needed — only a step inside the region with *sufficient reduction of the model*. The yardstick is the Cauchy point.

Algorithm 4.2

Solve the *linear* version of (4.3): $p_k^S = \arg \min_{\|p\| \leq \Delta_k} f_k + g_k^T p$ (4.9).

Minimize the model along it: $\tau_k = \arg \min_{\tau \geq 0} m_k(\tau p_k^S)$ s.t. $\|\tau p_k^S\| \leq \Delta_k$ (4.10).

Set $p_k^C = \tau_k p_k^S$.

Closed form: $p_k^S = -\frac{\Delta_k}{\|g_k\|} g_k$, and

$$p_k^C = -\tau_k \frac{\Delta_k}{\|g_k\|} g_k, \quad \tau_k = \begin{cases} 1 & \text{if } g_k^T B_k g_k \leq 0; \\ \min(\|g_k\|^3 / (\Delta_k g_k^T B_k g_k), 1) & \text{otherwise.} \end{cases} \quad (4.11-4.12)$$

- ▶ Nonpositive curvature along g : the model decreases forever along $-g$; go to the boundary.
- ▶ Positive curvature: 1-D convex quadratic in τ ; take its minimizer or the boundary, whichever is first.
- ▶ **No factorization required.** Cost: one matrix–vector product.

Why not stop at the Cauchy point?

The Cauchy point gives global convergence (§4.2) and costs almost nothing. So why look further?

Because it is steepest descent in disguise

Always taking p_k^C is the steepest descent method with a particular step-length rule. Chapter 3 showed what that means: linear convergence with rate $\left(\frac{\kappa-1}{\kappa+1}\right)^2$, hopeless on ill-conditioned problems.

The Cauchy point uses B_k only to set the *length* of the step. Rapid convergence needs B_k to shape the *direction* too — and needs B_k to carry valid curvature.

Design pattern for §4.1

Compute p^C ; then improve on it, in such a way that the **full step** $p^B = -B^{-1}g$ is chosen whenever $B \succ 0$ and $\|p^B\| \leq \Delta$. With $B = \nabla^2 f_k$ or a quasi-Newton approximation this recovers superlinear convergence (§4.4).

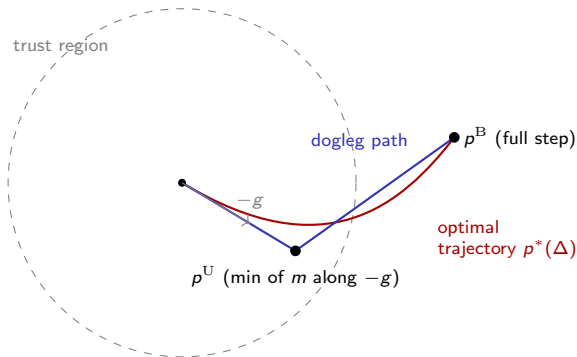
Three methods: **dogleg**, **two-dimensional subspace minimization**, and (Ch. 7) **CG–Steihaug**. All satisfy $m(p) \leq m(p^C)$.

The dogleg method: the exact trajectory (Figure 4.4)

Assume $B \succ 0$. How does the exact solution $p^*(\Delta)$ move as Δ grows?

$$p^*(\Delta) = p^B \quad \text{when } \Delta \geq \|p^B\| \quad (4.13), \quad p^*(\Delta) \approx -\Delta \frac{g}{\|g\|} \quad \text{when } \Delta \text{ is small} \quad (4.14).$$

In between it follows a curved path. The dogleg replaces the curve by **two line segments**.



The dogleg method: the path and why it works (Lemma 4.2)

First leg: origin to the minimizer of m along steepest descent,

$$p^U = -\frac{g^T g}{g^T B g} g. \quad (4.15)$$

Second leg: p^U to p^B . Parametrize $\tau \in [0, 2]$:

$$\tilde{p}(\tau) = \begin{cases} \tau p^U, & 0 \leq \tau \leq 1, \\ p^U + (\tau - 1)(p^B - p^U), & 1 \leq \tau \leq 2. \end{cases} \quad (4.16)$$

Choose p to minimize m along this path subject to $\|p\| \leq \Delta$.

Lemma 4.2 ($B \succ 0$)

(i) $\|\tilde{p}(\tau)\|$ is increasing in τ ; (ii) $m(\tilde{p}(\tau))$ is decreasing in τ .

Proof of (i) on the second leg: $h(\alpha) = \frac{1}{2}\|p^U + \alpha(p^B - p^U)\|^2$ has

$h'(\alpha) \geq -(p^U)^T(p^U - p^B) = g^T g \frac{g^T B^{-1} g}{g^T B g} \left[1 - \frac{(g^T g)^2}{(g^T B g)(g^T B^{-1} g)} \right] \geq 0$ by Cauchy-Schwarz. (ii) is a similar one-line computation.

Consequence. The path crosses the boundary $\|p\| = \Delta$ *exactly once* if $\|p^B\| \geq \Delta$, never otherwise. So: take p^B if it fits; else solve the scalar quadratic $\|p^U + (\tau - 1)(p^B - p^U)\|^2 = \Delta^2$ for τ . One factorization of B ; no iteration.

Dogleg with the exact Hessian — and a criticism

If $\nabla^2 f_k \succ 0$: set $B = \nabla^2 f_k$, p^B = Newton step, and dogleg gives the **Newton-dogleg** step. Near a solution satisfying second-order sufficient conditions this is just Newton's method — fast local convergence (§4.4).

If $\nabla^2 f_k$ is indefinite: replace it by a modified Hessian from §3.4 and proceed.

The book's own objection

A modified factorization perturbs the diagonal of $\nabla^2 f_k$ in a somewhat arbitrary way. But the trust region *already supplies a modification*: the exact solution is $(\nabla^2 f_k + \lambda I)^{-1} g_k$ with λ set by Δ_k (Theorem 4.1). Modifying first and trusting second is redundant, and the benefits of the trust region may be lost.

Verdict: Newton-dogleg is most appropriate for *convex* f . For the general case, use something that admits negative curvature directly.

(Extending dogleg to indefinite B is possible but pointless: p^B is then not the unconstrained minimizer of m , so the second leg has no meaning.)

Two-dimensional subspace minimization

Widen the dogleg's search from the two segments to the whole plane they span:

$$\min_p m(p) = f + g^T p + \frac{1}{2} p^T B p \quad \text{s.t. } \|p\| \leq \Delta, \quad p \in \text{span}[g, B^{-1}g]. \quad (4.17)$$

A problem in two variables — reduces to the roots of a quartic. p^C is feasible, so the reduction is at least the Cauchy reduction; the dogleg path lies in this plane, so it is at least as good as dogleg.

Indefinite B : the honest extension

With $\lambda_1 < 0$ the most negative eigenvalue, use instead

$$\text{span}[g, (B + \alpha I)^{-1}g], \quad \alpha \in (-\lambda_1, -2\lambda_1], \quad (4.18)$$

so $B + \alpha I \succ 0$ (the slack lets a Lanczos estimate of λ_1 suffice). If $\|(B + \alpha I)^{-1}g\| \leq \Delta$, take instead $p = -(B + \alpha I)^{-1}g + v$ with $v^T (B + \alpha I)^{-1}g \leq 0$ (4.19) — push to the boundary along a direction that includes negative curvature. Zero but no negative eigenvalues: $p = p^C$.

Reduction in m is often close to the exact solution's. Cost: **one factorization** of B or $B + \alpha I$, versus two or three for the nearly exact method of §4.3.

Preview: CG–Steihaug (Algorithm 7.2) — the large-scale route

When $B_k = \nabla^2 f_k$ is large and sparse, factorizations are off the table. Run the *conjugate gradient* method (Ch. 5) on m_k and stop early on three signals.

Algorithm 7.2 (CG–Steihaug)

Given $\epsilon_k > 0$. Set $z_0 = 0$, $r_0 = \nabla f_k$, $d_0 = -r_0$. If $\|r_0\| < \epsilon_k$ return $p_k = 0$.

for $j = 0, 1, 2, \dots$

if $d_j^T B_k d_j \leq 0$: **[negative curvature]** find τ with $p_k = z_j + \tau d_j$ minimizing m_k and $\|p_k\| = \Delta_k$; return.

$\alpha_j = r_j^T r_j / d_j^T B_k d_j$; $z_{j+1} = z_j + \alpha_j d_j$;

if $\|z_{j+1}\| \geq \Delta_k$: **[boundary]** find $\tau \geq 0$ with $p_k = z_j + \tau d_j$, $\|p_k\| = \Delta_k$; return.

$r_{j+1} = r_j + \alpha_j B_k d_j$; **if** $\|r_{j+1}\| < \epsilon_k$: **[converged]** return $p_k = z_{j+1}$.

$\beta_{j+1} = r_{j+1}^T r_{j+1} / r_j^T r_j$; $d_{j+1} = -r_{j+1} + \beta_{j+1} d_j$.

- ▶ First CG step is the Cauchy point, so the Cauchy reduction is guaranteed.
- ▶ Needs only $B_k d$ products — Hessian-free. Handles indefiniteness by construction.
- ▶ Near a good solution the bound goes inactive and this becomes inexact Newton (Ch. 7).

We will understand *why* the inner loop looks like this after Chapter 5.

The Cauchy reduction (Lemma 4.3, Theorem 4.4)

Everything in §4.1 was justified by “at least the Cauchy decrease”. Quantify it:

Lemma 4.3

$$m_k(0) - m_k(p_k^C) \geq \frac{1}{2} \|g_k\| \min \left(\Delta_k, \frac{\|g_k\|}{\|B_k\|} \right). \quad (4.21)$$

Proof: three cases from (4.12) — $g^T Bg \leq 0$; $g^T Bg > 0$ with $\tau < 1$; $g^T Bg > 0$ with $\tau = 1$ — each a two-line computation. (Homework.)

Theorem 4.4

If $\|p_k\| \leq \Delta_k$ and $m_k(0) - m_k(p_k) \geq c_2(m_k(0) - m_k(p_k^C))$, then

$$m_k(0) - m_k(p_k) \geq c_1 \|g_k\| \min \left(\Delta_k, \frac{\|g_k\|}{\|B_k\|} \right), \quad c_1 = c_2/2. \quad (4.20)$$

Dogleg, 2D subspace, and Steihaug all give $m(p) \leq m(p^C)$, hence (4.20) with $c_1 = \frac{1}{2}$.

Read (4.20) next to the Armijo condition: when Δ_k is the smaller term, the required model reduction is proportional to the gradient and to the step size. Same idea, different currency.

Convergence to stationary points (Theorems 4.5, 4.6)

Assumptions: $\|B_k\| \leq \beta$; f bounded below on the level set $S = \{x : f(x) \leq f(x_0)\}$ and Lipschitz continuously differentiable on a neighborhood $S(R_0)$; every p_k satisfies (4.20) and $\|p_k\| \leq \gamma \Delta_k$ for some $\gamma \geq 1$ (4.25).

Theorem 4.5 ($\eta = 0$)

$$\liminf_{k \rightarrow \infty} \|g_k\| = 0. \quad (4.26)$$

Theorem 4.6 ($\eta \in (0, \frac{1}{4})$)

$$\lim_{k \rightarrow \infty} g_k = 0. \quad (4.33)$$

The engine of both proofs

Taylor gives $|m_k(p_k) - f(x_k + p_k)| \leq (\beta/2 + \beta_1)\|p_k\|^2$ (4.27), so

$$|\rho_k - 1| = \frac{|m_k(p_k) - f(x_k + p_k)|}{m_k(0) - m_k(p_k)} \leq \frac{C\Delta_k^2}{c_1\|g_k\| \min(\Delta_k, \|g_k\|/\beta)}.$$

If $\|g_k\| \geq \epsilon$ and Δ_k is small, the right side is small: $\rho_k \rightarrow 1$ as the region shrinks, so the radius cannot collapse and steps keep being accepted. Each accepted step then reduces f by at least $\eta c_1 \epsilon \min(\Delta_k, \epsilon/\beta)$; infinitely many of them contradict f bounded below.

Same conclusion as Zoutendijk (stationary points only), by a different mechanism: there the Wolfe conditions bounded α_k below; here the model's accuracy at small Δ does.

Nearly exact solutions: turning Theorem 4.1 into an algorithm

For small n , use the model more fully. By (4.8a), the solution has the form

$$p(\lambda) = -(B + \lambda I)^{-1}g$$

for λ large enough that $B + \lambda I \succ 0$; we need $\lambda \geq 0$ with

$$\|p(\lambda)\| = \Delta. \quad (4.37)$$

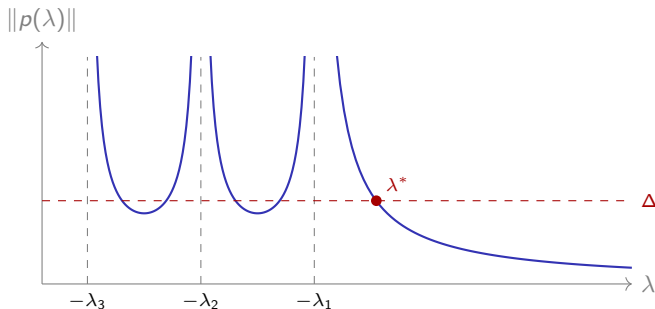
A **one-dimensional root-finding problem** in λ . (If $B \succ 0$ and $\|B^{-1}g\| \leq \Delta$, then $\lambda = 0$ and we are done.)

With $B = Q\Lambda Q^T$, $\lambda_1 \leq \dots \leq \lambda_n$:

$$p(\lambda) = -\sum_{j=1}^n \frac{q_j^T g}{\lambda_j + \lambda} q_j, \quad \|p(\lambda)\|^2 = \sum_{j=1}^n \frac{(q_j^T g)^2}{(\lambda_j + \lambda)^2}. \quad (4.38-4.39)$$

On $(-\lambda_1, \infty)$: continuous, nonincreasing, $\rightarrow 0$ as $\lambda \rightarrow \infty$ (4.40), and $\rightarrow \infty$ as $\lambda \downarrow -\lambda_j$ whenever $q_j^T g \neq 0$ (4.41). So if $q_1^T g \neq 0$ there is a **unique** $\lambda^* \in (-\lambda_1, \infty)$ with $\|p(\lambda^*)\| = \Delta$ — and it is the one satisfying (4.8c).

$\|p(\lambda)\|$ as a function of λ (Figure 4.5)



There may be other $\lambda < -\lambda_1$ with $\|p(\lambda)\| = \Delta$; they fail (4.8c) and are not solutions. The branch to the right of $-\lambda_1$ is the one that matters.

Root finding done right (Algorithm 4.3)

Newton on $\phi_1(\lambda) = \|p(\lambda)\| - \Delta$ (4.42) is a poor idea: near $-\lambda_1$, $\phi_1(\lambda) \approx C_1/(\lambda + \lambda_1) + C_2$, highly nonlinear.

Reformulate so the function is nearly *linear* near the root:

$$\phi_2(\lambda) = \frac{1}{\Delta} - \frac{1}{\|p(\lambda)\|} \approx \frac{1}{\Delta} - \frac{\lambda + \lambda_1}{C_3} \quad \text{near } -\lambda_1.$$

Newton on ϕ_2 , after some manipulation, becomes:

Algorithm 4.3 (Trust-Region Subproblem)

Given $\lambda^{(0)}$, $\Delta > 0$:

for $\ell = 0, 1, 2, \dots$

Factor $B + \lambda^{(\ell)}I = R^T R$; solve $R^T R p_\ell = -g$, $R^T q_\ell = p_\ell$;

$$\lambda^{(\ell+1)} = \lambda^{(\ell)} + \left(\frac{\|p_\ell\|}{\|q_\ell\|} \right)^2 \left(\frac{\|p_\ell\| - \Delta}{\Delta} \right) \quad (4.44)$$

- ▶ One Cholesky factorization per iteration — the main cost.
- ▶ Needs safeguards: if $\lambda^{(\ell)} < -\lambda_1$ the factorization does not exist.
- ▶ **Practical codes stop after two or three iterations**, well short of full accuracy.

The hard case

Everything above assumed $q_1^T g \neq 0$ (more precisely $Q_1^T g \neq 0$ for the eigenspace of λ_1). If $q_1^T g = 0$, the pole at $-\lambda_1$ disappears, and it can happen that

$$\|p(\lambda)\| < \Delta \quad \text{for all } \lambda \in (-\lambda_1, \infty)$$

— no root in the open interval (Figure 4.7). Theorem 4.1 still guarantees a solution, so the only possibility is $\lambda = -\lambda_1$.

At $\lambda = -\lambda_1$ the matrix $B - \lambda_1 I$ is singular; let z be a unit eigenvector for λ_1 . Then $q_j^T z = 0$ for $\lambda_j \neq \lambda_1$, and

$$p = \sum_{j: \lambda_j \neq \lambda_1} \frac{q_j^T g}{\lambda_j + \lambda} q_j + \tau z, \quad \|p\|^2 = \sum_{j: \lambda_j \neq \lambda_1} \frac{(q_j^T g)^2}{(\lambda_j + \lambda)^2} + \tau^2, \quad (4.45)$$

so τ can always be chosen to make $\|p\| = \Delta$, and (4.8) holds.

Meaning

The gradient has no component along the most negative curvature direction, so the model is indifferent to it at first order — and the solution is to *step along that direction anyway* until the boundary. This is how a trust-region method leaves a saddle point.

The payoff of near-exact solutions (Theorem 4.8)

Near-exact methods measure inexactness differently from §4.1:

$$m(0) - m(p) \geq c_1(m(0) - m(p^*)), \quad \|p\| \leq \gamma\Delta. \quad (4.52)$$

Compare (4.20): at a **saddle point** ($g = 0$, B with negative eigenvalues) the right side of (4.20) is zero — dogleg-type methods *stop there*. The right side of (4.52) is positive — the algorithm is forced to move.

Theorem 4.8 ($B_k = \nabla^2 f_k$, steps satisfying (4.52))

Under the assumptions of Theorem 4.6 with $f \in C^2$ on S : $\|g_k\| \rightarrow 0$. If moreover S is compact, then either the algorithm terminates at a point satisfying the *second-order necessary conditions*, or $\{x_k\}$ has a limit point in S at which they hold.

This is more than Chapter 3 could promise

Every line-search result ended at “stationary point”. With the exact Hessian and a near-exact subproblem solution, trust-region methods converge to points where $\nabla^2 f \succeq 0$ as well — they *do not get stuck at saddles*. The price: the extra factorizations of Algorithm 4.3, and $B_k = \nabla^2 f_k$ exactly.

Local convergence of trust-region Newton (Theorem 4.9)

The fast Newton rate requires that the bound $\|p\| \leq \Delta_k$ eventually *stop interfering*: near x^* the subproblem solution should sit well inside the region and approach the Newton step. Steps with this property are **asymptotically similar to Newton steps**.

Theorem 4.9

Let f be twice Lipschitz continuously differentiable near x^* satisfying second-order sufficient conditions, and $x_k \rightarrow x^*$. Suppose that for all large k the algorithm uses $B_k = \nabla^2 f_k$, its steps satisfy the Cauchy criterion (4.20), and whenever $\|p_k^N\| \leq \frac{1}{2}\Delta_k$,

$$\|p_k - p_k^N\| = o(\|p_k^N\|). \quad (4.53)$$

Then Δ_k becomes inactive for all large k , and $x_k \rightarrow x^*$ superlinearly.

Shape of the proof. (4.53) gives $\|p_k\| \leq 2\|p_k^N\| \leq 2\|\nabla^2 f_k^{-1}\|\|g_k\|$; with (4.20) this yields $\rho_k \rightarrow 1$, so the radius is eventually never reduced and $\|p_k^N\| \leq \frac{1}{2}\Delta_k$ holds from some point on. Then p_k is the near-Newton step, and Theorem 3.5 plus (4.53) give the rate.

If $p_k = p_k^N$ exactly for large k : **quadratic**. Dogleg, 2D subspace, and Algorithm 4.3 (with a check of p_k^N first) all do this — p_k^N lies on the dogleg path, in the subspace, and inside the region, and is the unconstrained minimizer of m_k there.

Scaling: elliptical trust regions

A spherical region is wrong for a poorly scaled f : the model is trustworthy far along insensitive directions and only a short way along sensitive ones. Confidence should be roughly uniform on the boundary — so make the boundary an *ellipse*:

$$\min_p m_k(p) \quad \text{s.t. } \|Dp\| \leq \Delta_k, \quad (4.57)$$

D diagonal, d_{ii} large for sensitive components (from $\partial^2 f / \partial x_i^2$, say). D may change per iteration as long as $d_{ii} \in [d_{lo}, d_{hi}]$.

Generalized Cauchy point (Algorithm 4.4)

$$p_k^S = -\frac{\Delta_k}{\|D^{-1}g_k\|} D^{-2}g_k, \quad \tau_k = \begin{cases} 1 & g_k^T D^{-2}B_k D^{-2}g_k \leq 0 \\ \min(\|D^{-1}g_k\|^3 / (\Delta_k g_k^T D^{-2}B_k D^{-2}g_k), 1) & \text{else} \end{cases} \quad (4.60-4.61)$$

Simpler: substitute $\tilde{p} = Dp$ and run everything in the scaled variables — $g_k \rightarrow D^{-1}g_k$, $B_k \rightarrow D^{-1}B_k D^{-1}$, region spherical again.

Other norms. $\|p\|_1 \leq \Delta$ or $\|p\|_\infty \leq \Delta$ (a box) are also used; the box is natural when bound constraints $l \leq x \leq u$ are present, since the intersection of a box with a box is a box (Ch. 16–17).

Chapter 4 in one table

Subproblem solver	Needs	Cost / iter	Indefinite B ?	Newton step recovered?
Cauchy point	Bg	1 mat-vec	yes	no (steepest descent)
Dogleg	$B \succ 0$	1 factorization	no	yes
2D subspace	λ_1 est.	1 factorization	yes (4.18)	yes
Nearly exact (Alg. 4.3)	Cholesky	2–3 factorizations	yes, incl. hard case	yes
CG–Steihaug (Alg. 7.2)	Bd products	few mat-vecs	yes	yes (inexact)

Result	Says
Theorem 4.1	exact solution = $(B + \lambda I)^{-1}(-g)$, λ set by Δ
Lemma 4.3 / Thm 4.4	Cauchy fraction $\Rightarrow$ model reduction bound (4.20)
Theorems 4.5, 4.6	(4.20) + radius logic $\Rightarrow \ g_k\ \rightarrow 0$
Theorem 4.8	exact Hessian + near-exact steps $\Rightarrow$ second-order points
Theorem 4.9	bound goes inactive $\Rightarrow$ superlinear / quadratic

Line search vs. trust region — where each lives from here on

Line search

- ▶ Indefinite Hessian must be *repaired* first (§3.4)
- ▶ Converges to stationary points
- ▶ Natural for quasi-Newton (Wolfe $\Rightarrow s^T y > 0$)
- ▶ Used in: BFGS / L-BFGS (Ch. 6–7), SQP (Ch. 18), IPOPT's filter line search (Ch. 19)

Trust region

- ▶ Indefinite Hessian handled *by the constraint*
- ▶ Can converge to second-order points (Thm 4.8)
- ▶ Natural for exact Hessians and for nonlinear least squares
- ▶ Used in: Levenberg–Marquardt (Ch. 10), nonlinear equations (Ch. 11), SR1 (Ch. 6), trust-region SQP and IPM variants (Ch. 18–19)

One thing to remember

The trust-region step is $-(B + \lambda I)^{-1}g$ with λ chosen *by the radius*. Levenberg–Marquardt, Hessian modification by τI , and inertia correction in interior-point methods are all this same object under different names.

Homework

Derivations

- 1 Derive the closed form (4.11)–(4.12) of the Cauchy point from Algorithm 4.2.
- 2 Complete the proof of Lemma 4.2(i): show the bracketed factor is nonnegative by Cauchy–Schwarz, and prove (ii).
- 3 Prove Lemma 4.3 (all three cases).
- 4 Using (4.39), show that $\phi_2(\lambda) = 1/\Delta - 1/\|p(\lambda)\|$ is approximately linear for λ slightly larger than $-\lambda_1$, and derive the update (4.44) from Newton's method on ϕ_2 .
- 5 Verify that p in (4.45) with $\lambda = -\lambda_1$ satisfies all three conditions (4.8).

Implementation

6. Implement Algorithm 4.1 with the dogleg subproblem solver and $B_k = \nabla^2 f_k$. Run on Rosenbrock from $(-1.2, 1)$. Log ρ_k , Δ_k , and whether the step hit the boundary. Compare iteration counts with your Chapter 3 Newton line search.
7. Implement Algorithm 4.3 (two or three iterations, with a $\lambda > -\lambda_1$ safeguard) for $n = 2$. Construct an instance of the hard case — B indefinite with $g \perp q_1$ — and confirm that (4.45) is needed.
8. (After Ch. 5) Implement CG–Steihaug and rerun item 6. Count Hessian–vector products.

Next: Ch. 5 — *conjugate gradients: the engine inside Steihaug and Newton-CG*.